

OpenStax Calculus Volume 2

Section 2.2 Exercises: Determining Volumes by Slicing

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1. Derive the formula for the volume of a sphere using the slicing method.
2. Use the slicing method to derive the formula for the volume of a cone.
3. Use the slicing method to derive the formula for the volume of a tetrahedron with side length a .
4. Use the disk method to derive the formula for the volume of a trapezoidal cylinder.
5. Explain when you would use the disk method versus the washer method. When are they interchangeable?

For the following exercises, draw a typical slice and find the volume using the slicing method for the given volume.

6. A pyramid with height 6 units and square base of side 2 units, as pictured here.
[Figure: This figure is a pyramid with base width of 2 and height of 6 units.]
7. A pyramid with height 4 units and a rectangular base with length 2 units and width 3 units, as pictured here.
[Figure: This figure is a pyramid with base width of 2, length of 3, and height of 4 units.]
8. A tetrahedron with a base side of 4 units, as seen here.
[Figure: This figure is an equilateral triangle with side length of 4 units.]
9. A pyramid with height 5 units, and an isosceles triangular base with lengths of 6 units and 8 units, as seen here.
[Figure: This figure is a pyramid with a triangular base. The view is of the base. The sides of the triangle measure 6 units, 8 units, and 8 units. The height of the pyramid is 5 units.]
10. A cone of radius r and height h has a smaller cone of radius $r/2$ and height $h/2$ removed from the top, as seen here. The resulting solid is called a frustum.
[Figure: This figure is a 3-dimensional graph of an upside down cone. The cone is inside of a rectangular prism that represents the xyz coordinate system. The radius of the bottom of the cone is “ r ” and the radius of the top of the cone is labeled “ $r/2$ ”.]

For the following exercises, draw an outline of the solid and find the volume using the slicing method.

11. The base is a circle of radius a . The slices perpendicular to the base are squares.
12. The base is a triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Slices perpendicular to the x -axis are semicircles.
13. The base is the region under the parabola $y = 1 - x^2$ in the first quadrant. Slices perpendicular to the xy -plane and parallel to the y -axis are squares.
14. The base is the region under the parabola $y = 1 - x^2$ and above the x -axis. Slices perpendicular to the y -axis are squares.
15. The base is the region enclosed by $y = x^2$ and $y = 9$. Slices perpendicular to the x -axis are right isosceles triangles. The intersection of one of these slices and the base is the leg of the triangle.
16. The base is the area between $y = x$ and $y = x^2$. Slices perpendicular to the x -axis are semicircles.

For the following exercises, draw the region bounded by the curves. Then, use the disk method to find the volume when the region is rotated around the x -axis.

17. $x + y = 8$, $x = 0$, and $y = 0$
18. $y = 2x^2$, $x = 0$, $x = 4$, and $y = 0$
19. $y = e^x + 1$, $x = 0$, $x = 1$, and $y = 0$
20. $y = x^4$, $x = 0$, and $y = 1$ for $x \geq 0$
21. $y = \sqrt{x}$, $x = 0$, $x = 4$, and $y = 0$
22. $y = \sin x$, $y = \cos x$, and $x = 0$
23. $y = \frac{1}{x}$, $x = 2$, and $y = 3$
24. $x^2 - y^2 = 9$ and $x + y = 9$, $y = 0$ and $x = 0$

For the following exercises, draw the region bounded by the curves. Then, find the volume when the region is rotated around the y -axis.

25. $y = 4 - \frac{1}{2}x$, $x = 0$, and $y = 0$
26. $y = 2x^3$, $x = 0$, $x = 1$, and $y = 0$
27. $y = 3x^2$, $x = 0$, and $y = 3$
28. $y = \sqrt{4 - x^2}$, $y = 0$, and $x = 0$
29. $y = \frac{1}{\sqrt{x+1}}$, $x = 0$, $x = 3$, and $y = 0$
30. $x = \sec(y)$ and $y = \frac{\pi}{4}$, $y = 0$ and $x = 0$
31. $y = \frac{1}{x+1}$, $x = 0$, $x = 2$, and $y = 0$
32. $y = 4 - x$, $y = x$, and $x = 0$

For the following exercises, draw the region bounded by the curves. Then, find the volume when the region is rotated around the x -axis.

33. $y = x + 2$, $y = x + 6$, $x = 0$, and $x = 5$
34. $y = x^2$ and $y = x + 2$
35. $x^2 = y^3$ and $x^3 = y^2$
36. $y = 4 - x^2$ and $y = 2 - x$
37. [T] $y = \cos x$, $y = e^{-x}$, $x = 0$, and $x = 1.2927$
38. $y = \sqrt{x}$ and $y = x^2$
39. $y = \sin x$, $y = 5 \sin x$, $x = 0$ and $x = \pi$
40. $y = \sqrt{1 + x^2}$ and $y = \sqrt{4 - x^2}$

For the following exercises, draw the region bounded by the curves. Then, use the washer method to find the volume when the region is revolved around the y -axis.

41. $y = \sqrt{x}$, $x = 4$, and $y = 0$
42. $y = x + 2$, $y = 2x - 1$, and $x = 0$
43. $y = \sqrt[3]{x}$ and $y = x^3$
44. $x = e^{2y}$, $x = y^2$, $y = 0$, and $y = \ln(2)$
45. $x = \sqrt{9 - y^2}$, $x = e^{-y}$, $y = 0$, and $y = 3$
46. Yogurt containers can be shaped like frustums. Rotate the line $y = \frac{1}{m}x$ around the y -axis to find the volume between $y = a$ and $y = b$.
[Figure: This figure has two parts. The first part is a solid cone. The base of the cone is wider than the top. It is shown in a 3-dimensional box. Underneath the cone is an image of a yogurt container with the same shape as the figure.]
47. Rotate the ellipse $(x^2/a^2) + (y^2/b^2) = 1$ around the x -axis to approximate the volume of a football, as seen here.
[Figure: This figure has an oval that is approximately equal to the image of a football.]
48. Rotate the ellipse $(x^2/a^2) + (y^2/b^2) = 1$ around the y -axis to approximate the volume of a football.
49. A better approximation of the volume of a football is given by the solid that comes from rotating $y = \sin x$ around the x -axis from $x = 0$ to $x = \pi$. What is the volume of this football approximation, as seen here?
[Figure: This figure has a 3-dimensional oval shape. It is inside of a box parallel to the x -axis on the bottom front edge of the box. The y -axis is vertical to the solid.]
50. What is the volume of the Bundt cake that comes from rotating $y = \sin x$ around the y -axis from $x = 0$ to $x = \pi$?
[Figure: This figure is a graph of a 3-dimensional solid. It is round, bigger towards the bottom. It has a hole in the center that progressively gets smaller towards the bottom. Next to the graph is an image of a bundt cake, resembling the solid.]

For the following exercises, find the volume of the solid described.

51. The base is the region between $y = x$ and $y = x^2$. Slices perpendicular to the x -axis are semicircles.
52. The base is the region enclosed by the generic ellipse $(x^2/a^2) + (y^2/b^2) = 1$. Slices perpendicular to the x -axis are semicircles.
53. Bore a hole of radius a down the axis of a right cone of height b and radius b through the base of the cone as seen here.
[Figure: This figure is an upside down cone. It has a radius of the top as "b", center at "a", and height as "b".]
54. Find the volume common to two spheres of radius r with centers that are $2h$ apart, as shown here.
[Figure: This figure has two circles that intersect. Both circles have radius "r". There is a line segment from one center to the other. In the middle of the intersection of the circles is point "h". It is on the line segment.]
55. Find the volume of a spherical cap of height h and radius r where $h < r$, as seen here.
[Figure: This figure a portion of a sphere. This spherical cap has radius "r" and height "h".]
56. Find the volume of a sphere of radius R with a cap of height h removed from the top, as seen here.
[Figure: This figure is a sphere with a top portion removed. The radius of the sphere is "R". The distance from the center to where the top portion is removed is "R-h".]